

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0229

Hato Rey Gis 1403-1404 100 kV
June 22, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0229. The Interconnection Customer proposes a 106.8 MW solar photovoltaic facility interconnecting at the Hato Rey Gis 1403-1404 100 kV substation in the Indiana area. The request was received on April 19, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 174 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,495,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Madeline to East Methuen 230 kV
Contingency: Baraga Vlg to Gray'S Flat 230 kV L
Loading: 116.4% (Rating: 607 MVA)
DFAX: 10.1%
Pre-project loading: 114.6%
2. Facility: Myrtle Point to Galion Generating Station 230 kV
Contingency: Trego to Beaver 500 kV
Loading: 111.0% (Rating: 628 MVA)
DFAX: 6.5%
Pre-project loading: 109.9%
3. Facility: Eufaula 230 kV/1 kV TX
Contingency: Imperial to Eufaula 500 kV Line
Loading: 108.6% (Rating: 1,126 MVA)
DFAX: 12.0%
Pre-project loading: 107.5%
4. Facility: Don to Hato Ry Gis 1403-1404 100 kV
Contingency: Edes Tap 1 to Hato Rey Gis 1403-1404 100 kV
Loading: 108.5% (Rating: 87 MVA)
DFAX: 100.0%
Pre-project loading: 14.2%
5. Facility: Moreno Tap to Kampsville 230 kV
Contingency: Trego to Swan Valley 500 kV
Loading: 107.4% (Rating: 570 MVA)
DFAX: 7.9%
Pre-project loading: 105.9%

6. Facility: Moreno Tap Substation to Eastown Rd 230 kV
Contingency: Swan Valley Substation to Trego 500 kV
Loading: 107.2% (Rating: 592 MVA)
DFAX: 7.9%
Pre-project loading: 105.8%
7. Facility: Ft Humbug 500/345kV Transformer
Contingency: Orchard Rd to Ft Humbug 500 kV
Loading: 106.9% (Rating: 1,123 MVA)
DFAX: 5.2%
Pre-project loading: 106.4%
8. Facility: Zona Libre 6007 to Kampsville 230 kV
Contingency: Swan Vly to Trego 500 kV
Loading: 106.8% (Rating: 562 MVA)
DFAX: 7.9%
Pre-project loading: 105.3%
9. Facility: George Neal S to Zona Libre 6007 230 kV line
Contingency: Swan Alley to Trego 500 kV
Loading: 106.5% (Rating: 543 MVA)
DFAX: 7.9%
Pre-project loading: 104.9%
10. Facility: Myrtle Point to George Neal S 230 kV Line
Contingency: Swan Vly to Trego Substation 500 kV
Loading: 105.5% (Rating: 528 MVA)
DFAX: 7.9%
Pre-project loading: 103.9%
11. Facility: Norwich to 15Th Avenue South 230 kV line
Contingency: Omni-Hackbery to Norwich 500 kV
Loading: 104.8% (Rating: 507 MVA)
DFAX: 6.3%
Pre-project loading: 103.4%
12. Facility: Montour Sta 500/345kV Transformer
Contingency: Westwood Generation Llc to Coronet 500 kV
Loading: 104.2% (Rating: 883 MVA)
DFAX: 5.0%
Pre-project loading: 103.6%
13. Facility: 15Th Avenue S to Lucent 230 kV
Contingency: Omni-Hackberry to Norwih 500 kV
Loading: 103.9% (Rating: 499 MVA)
DFAX: 6.3%
Pre-project loading: 102.5%
14. Facility: Little Chute Substation to Ft Humbug Substation 345 kV
Contingency: Fort Humbug to Coronet 500 kV Line
Loading: 102.2% (Rating: 1,440 MVA)
DFAX: 5.7%
Pre-project loading: 101.8%
15. Facility: Hilltop to Strawberry Belt 345 kV
Contingency: Hilltop to Talenenergy Martins Crk Llc Williamsport 500 kV line
Loading: 102.1% (Rating: 1,135 MVA)
DFAX: 6.5%
Pre-project loading: 101.4%
16. Facility: Baraga Vlg to Gray'S Flat 230 kV
Contingency: Madeline to East Methuen 230 kV

Loading: 102.0% (Rating: 673 MVA)

DFAX: 9.6%

Pre-project loading: 100.4%

17. Facility: Eufaula 500/1 kV Autotransformer

Contingency: Imperial to Eufaula 500 kV

Loading: 101.9% (Rating: 1,201 MVA)

DFAX: 12.0%

Pre-project loading: 100.8%

18. Facility: Carbon Cr 500 kV/345 kV Auto

Contingency: Blackrock to Shawville Sta 345 kV

Loading: 100.9% (Rating: 983 MVA)

DFAX: 7.3%

Pre-project loading: 100.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
East Methuen to Madeline 230 kV	Rebuild line with higher-capacity conductor	\$37,344,000	1% (\$524,000)
Myrtle Pt Substation to Galion Generating Sta Substation 230 kV Line	Rebuild line with higher-capacity conductor	\$20,084,000	2% (\$348,000)
Eufaula 230-1 kV Autotransformer	Replace transformer with higher-rated unit	\$7,105,000	2% (\$128,000)
Don to Hato Ry Gis 1403-1404 100 kV	Reconductor line	\$723,000	100% (\$723,000)
Moreno Tap to Kampsville 230 kV L	Reconductor line	\$10,521,000	4% (\$379,000)
Eastown Road to Moreno Ap 230 kV	Reconductor line	\$16,626,000	4% (\$599,000)
Ft Humbug 500/345kV Transformer	Replace transformer with higher-rated unit	\$34,031,000	10% (\$3,381,000)
Zona Libre 6007 to Kampsville 230 kV	Reconductor line	\$10,502,000	4% (\$378,000)
George Neal S to Zona Libre 6007 230 kV line	Reconductor line	\$8,203,000	4% (\$295,000)
Gorge Neal S to Myrtle Point 230 kV	Reconductor line	\$4,420,000	4% (\$159,000)
15Th Avenue S to Norwich 230 kV	Reconductor line	\$4,841,000	2% (\$80,000)
Montour Sta Substation 500/345kV Auto	Replace transformer with higher-rated unit	\$21,071,000	10% (\$2,092,000)
15Th Avenue S to Lucent 230 kV	Reconductor line	\$4,095,000	2% (\$68,000)
Little Chute to Ft Humbug Substation 345 kV	Reconductor line	\$39,933,000	2% (\$889,000)
Hilltop to Strawberry Belt 345 kV	Reconductor line	\$11,781,000	2% (\$177,000)
Baraga Vlg to Gray'S Flat 230 kV	Reconductor line	\$5,935,000	2% (\$93,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Eufaula 500/1 kV Autotransformer	Replace transformer with higher-rated unit	\$38,863,000	2% (\$701,000)
Carbon Ctr 500/345 kV Transformer	Replace transformer with higher-rated unit	\$32,116,000	2% (\$481,000)
	Total allocated to project		\$11,495,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.